

0001 — Pullback Scale-In on the MC CC Setup — 2026-06-24

****Series:**** MC Setup Research Notes · Note 0001 ****Confidence:**** High — 5 years, full-cost, OOS-tested by year and stress-tested on exit assumptions; the NO-GO holds throughout.

TL;DR: Adding a second contract when price pulls back toward your stop, then targeting the original 1R, **is not worth it on the MC setup**. Across every test — first-principles math, tick-by-tick path analysis on 5 years of data, per-setup and time-of-day and context filters, a full pullback-depth sweep (0.25→1.0R), and an execution stress test — the add never clears the bar it needs to. Its apparent profit is an accounting artifact of marking unresolved trades at the session close; remove that one optimistic assumption and the edge disappears at every depth. **Recommendation: take the clean single-leg trade to 1R and skip the scale-in.** The only durable, real finding is a negative one: pullbacks in the afternoon (after 13:00 CT) are structurally bad and should be avoided regardless.

1. The setup (so this note stands alone)

The MC CC trade. The "MC" indicator fires a breakout signal (one of five subtypes, **CC1–CC5**). You enter at the signal bar's close (**E1**), place a stop a fixed distance away, and target **+1R** (1R = the entry-to-stop distance). On ES, 1 point = \$50/contract; a 10-point stop = **1R = \$500**.

The scale-in variant we're testing. Instead of one contract, you add a **second contract (leg 2)** if price first pulls back toward your stop by some fraction of R — e.g. a **50% pullback** means leg 2 fills halfway between entry and stop. Both contracts then exit at the original 1R target. The idea: leg 2's cheaper entry boosts the winner.

The three outcomes of any scale-in attempt (a long; shorts mirror), with p = pullback depth:

Outcome	Leg 1	Leg 2	Total	At $p=0.50$
Clean win — runs to target, never pulls back	+1R	(never added)	+1R	+1R
Scale-in win — pulls back, leg 2 fills, both reach 1R	+1R	$+(1+p)R$	$+(2+p)R$	+2.5R
Scale-in loss — pulls back, leg 2 fills, both stop	-1R	$-(1-p)R$	$-(2-p)R$	-1.5R

Cost model. \$5 round-turn commission + 1 tick (\$12.50) slippage **per leg** = \$17.50/leg. The scale-in pays it twice.

2. The question

Does the pullback scale-in add money over just taking the clean single-leg trade to 1R — and if so, at what pullback depth, in what conditions?

3. How we tested it (the menu)

1. **First-principles math** — what win rate does the add need to break even? 2. **Tick-by-tick path study** — what happened to all 5,580 signals over 5 years. 3. **Per-setup (CC) breakdown** — does the add help any subtype? 4. **Time-of-day** — does session phase change the picture? 5. **"Tell" hunt** — can any pre-trade signal pick the winners? 6. **Year-by-year validation** — does anything good survive out of sample? 7. **Pullback-depth sweep** — is 50% even the right depth (0.25 → 1.0R)? 8. **Execution stress test** — how much depends on optimistic exit assumptions?

Data: ba_signals_mc.parquet (5,580 signals, 2021-06 → 2026-06), replayed on real continuous ticks day-by-day. 5,444–5,490 have tick coverage. All P&L net of costs.

4. Results

4.1 The math — the add needs a win rate it doesn't have

Per-trade outcomes (\$500 R, full cost, 50% PB):

Outcome	Contracts	Net \$	Net R
Clean win	1	+\$482.50	+0.97R
Scale-in win	2	+\$1,215	+2.43R
Scale-in loss	2	-\$785	-1.57R

Breakeven win rates by structure (1R target):

Structure	Win	Loss	Breakeven win%
No scale-in (clean 1 contract)	+\$482.50	-\$517.50	51.8%
Scale-in @ 50%, 1R target	+\$1,215	-\$785	39.3%
Scale-in @ 50%, breakeven exit (the trap)	+\$240	-\$785	76.0%
Leg 2 alone (the marginal add decision)	+\$732.50	-\$267.50	26.8%

Read: on its own, the second contract only needs to reach 1R on **~27%** of the pullbacks it joins to be worth adding. Hold that 27% — it's the bar for everything below. (The marginal bar widens as the pullback gets shallower: it equals $(1-p)/2$, i.e. 37.5% at 0.25R, 33.5% at 0.33R, 25% at 0.50R, 17% at 0.66R.)

4.2 The path study — what actually happened (50% pullback, N=5,490)

Outcome	Count	% of all
Clean win (straight to 1R)	1,684	30.7%
Scale-in win (PB → 1R)	538	9.8%
Scale-in loss (PB → stop)	1,829	33.3%
Clean EOD (no pullback, no target)	789	14.4%
Pullback then EOD (unresolved)	604	11.0%

- **54% of signals pull back** to the 50% level, but only **18% of those then reach**

1R (538 / 2,971). The biggest single outcome of any signal is PB → stop (33%) — the pullback usually precedes a loss, not a bounce.

- Among resolved pullbacks (1R vs stop), the win rate is **22.7%** — below the ~27% the add needs. **On its own bet, the second contract loses.**

- Full population: scale-in net **\$306,256** vs single-leg **\$268,868** → the add appears to add **+\$37,388**. §4.8 shows that's a marking artifact.

4.3 Per-setup (CC) — no subtype rescues it

50% PB, full cost. SI = scale-in, SL = single-leg.

CC	N	PB→1R (% all)	PB-touch%	SI net	SL net	Add value	SI expR	SI PF
CC1	132	5.3%	47.0%	\$23,185	\$25,108	-\$1,922	0.061	1.39
CC2	906	10.7%	51.5%	\$65,260	\$52,170	+\$13,090	0.049	1.16
CC3	1,633	11.0%	57.5%	\$61,930	\$44,612	+\$17,318	0.020	1.08
CC4	1,656	10.0%	55.0%	\$80,810	\$72,685	+\$8,125	0.058	1.10
CC5	1,163	7.7%	51.0%	\$75,071	\$74,292	+\$779	0.073	1.15

The add is negative for CC1, ~zero for CC5 (the strongest standalone setup), and only meaningfully positive for CC2/CC3 — and §4.8 dissolves even those.

4.4 Time-of-day — the one strong pattern (CME Central Time)

Phase	N	Clean win	PB-touch	PB→1R (of all)	Resolved win%
Open (08:30–11:30)	2,338	33.7%	60.7%	12.4%	23.8%
Mid (11:30–13:00)	1,189	37.1%	57.3%	12.7%	26.7%
Late (13:00–14:45)	1,443	29.0%	54.3%	6.7%	17.5%
Close (14:45–15:15)	575	6.4%	15.3%	0.0%	—

The whole setup decays through the session. **Open+Mid = 24.7%** resolved (N=1,782) vs **Late+Close = 16.6%** (N=585). At 50% depth the add makes **+\$76,924 in Open+Mid** but loses **-\$39,535 in Late+Close** — netting the +\$37k above. The afternoon eats more than half the morning's add.

4.5 The "tell" hunt — full feature ranking (resolved binary, base 22.7%, N=2,367)

Every cheap pre-trade feature, best-to-worst bucket. <<< = ≥6 pts over base, (low) = ≥6 under. Causal (look-ahead-safe) tags via tag_signals.

Feature	Bucket (win% / N), best → worst
ext_ema (entry vs EMA20, trade dir)	far-pullback-side 30.5/174 <<< · below 27.1/59 · ≈EMA 25.0/76 · above 22.8/149 · chasing-extended 21.8/1909
session phase	Mid 26.7/565 · Open 23.8/1217 · Late 17.5/555 (low)
balance_state	True 25.9/437 · False 22.0/1930
dir_streak (consecutive same-dir)	4+ 25.9/555 · 1 22.6/957 · 2 20.9/512 · 3 20.7/343
AID_DirMatch (Always-In)	against-regime 25.7/451 · with-regime 22.0/1916
prior_ER (prior-day trend)	≥.5 26.2/515 · ≤.2 23.5/729 · .2-.35 21.3/550 · .35-.5 20.6/472
ER_intra_6	.2-.35 25.3/344 · .35-.5 25.2/408 · ≤.2 24.8/302 · ≥.5 20.8/1303
ER_intra_12	≤.2 24.4/746 · .35-.5 23.2/453 · .2-.35 22.1/485 · ≥.5 20.9/670
is_deep_pullback	True 24.3/515 · False 22.3/1852
AID_State	up +1 23.9/1234 · down -1 21.4/1133
dow	Wed 27.3/477 · Mon 24.8/408 · Tue 20.9/487 · Fri 20.7/492 · Thu 20.5/503
SignalType	CC2 24.2/401 · CC4 23.0/718 · CC3 22.9/780 · CC5 21.3/423 · CC1 15.6/45
is_long	long 23.5/1255 · short 21.9/1112
AID_bars since flip	1-3 25.0/541 · 11+ 22.6/1082 · 4-10 21.4/401 · on-flip 21.3/343
ext_vwap	>1σ 23.4/1423 · -1..0 22.3/341 · <-1σ 20.0/110 · 0..1σ 20.0/380
prior_inside_day	False 23.1/2156 · True 19.4/211
prior_adr_ext (prior trend day)	False 23.3/2033 · True 19.5/334

Best 2-feature combos clearing 30% (N≥60): Mid+balance 37.5/72 · Mid+CC4 31.5/178 · CC2+against-regime 31.0/100 · Mid+AID-1-3-bars 30.9/123 · CC5+balance 30.4/79.

All single tells are weak (3–8 pts). Two notes: the streak only helps at **4+** (2–3 are worse than singletons); Always-In points

counter-trend — using it as continuation confirmation hurts (consistent with its prior shelving as a gate). The 30%+ combos sit on N=70–130 with ± 11 -pt confidence bands overlapping the base — **overfit, not signal**.

4.6 Year-by-year — only the negative survives

Resolved win% (1R vs stop) by year × phase:

Year	Open	Mid	Late	Open+Mid	Late+Close
2021	23.5 (119)	24.2 (62)	18.3 (71)	23.8	17.3
2022	22.7 (247)	21.6 (125)	22.2 (144)	22.3	21.9
2023	29.0 (259)	29.5 (105)	12.0 (108)	29.1	11.3
2024	19.6 (240)	30.7 (101)	12.0 (100)	22.9	11.4
2025	23.7 (249)	31.3 (131)	20.8 (96)	26.3	18.9
2026	24.3 (103)	14.6 (41)	19.4 (36)	21.5	18.4

Whole-population PB→1R share (of all signals) by year × phase:

Year	Open+Mid	Late+Close
2021	11.8% (363)	5.3% (245)
2022	11.2% (739)	6.9% (465)
2023	15.9% (665)	3.1% (418)
2024	11.9% (656)	3.7% (328)
2025	13.0% (770)	5.2% (381)
2026	9.3% (334)	3.9% (181)

Robust: Late+Close is worse in **6/6 years** (morning pullbacks reach 1R ~2× more often). **Fragile:** "Open+Mid is a positive edge" clears the ~27% bar in only **2 of 6 years** (2023, 2025); 2022 had no phase edge at all. "Mid is the sweet spot" is a 2023–25 artifact (great those years, weak/noise 2021/22/26) — does not generalize.

4.7 Pullback-depth sweep (0.25 → 1.0R) — shallower looks better... but see §4.8

All signals (single-leg baseline net = \$268,868), exits marked at session close:

PB depth	Touch%	PB→1R (all)	Add hit% (1R/touch)	Bar to beat	PB→stop%	SI net	Add value
0.25R	71.5%	20.8%	29.1%	37.5%	33.6%	\$371,812	+\$102,945
0.33R	64.6%	16.2%	25.1%	33.5%	33.6%	\$329,227	+\$60,360
0.50R	54.6%	9.9%	18.1%	25.0%	33.6%	\$306,256	+\$37,389
0.66R	46.4%	5.2%	11.3%	17.0%	33.6%	\$238,047	–\$30,821
0.75R	42.6%	3.3%	7.7%	12.5%	33.6%	\$230,041	–\$38,826
1.00R	33.6%	0.0%	0.0%	0.0%	33.6%	\$236,860	–\$32,008

Add value (\$) by CC × depth:

CC	0.25	0.33	0.50	0.66	0.75	1.00
CC1	+9,546	+7,848	–1,922	–5,228	–4,634	–665
CC2	+36,284	+33,126	+13,090	–5,884	–3,701	–5,320
CC3	+26,568	+15,342	+17,318	–3,373	–859	–10,518
CC4	+6,625	+1,365	+8,125	–17,829	–35,597	–9,678
CC5	+23,922	+2,679	+779	+1,493	+5,964	–5,828

Add hit-rate (PB→1R / touched) by session × depth:

Phase	0.25	0.33	0.50	0.66	0.75
Open+Mid	33.8%	29.3%	21.0%	13.4%	9.3%
Late+Close	19.4%	15.7%	11.1%	5.9%	3.7%

Add value (\$) by session × depth:

Phase	0.25	0.33	0.50	0.66	0.75	1.00
Open+Mid	+134,012	+109,518	+76,924	+16,701	–2,047	–23,468
Late+Close	–31,068	–49,159	–39,535	–47,521	–36,779	–8,540

Add value falls monotonically with depth and goes negative beyond ~0.5R. **Red flag:** at every depth the add's hit rate is **below the bar it needs** (29.1% vs 37.5% at 0.25R). The directional bet loses at all depths — the positive numbers must come from somewhere other than reaching the target.

4.8 Execution stress test — the kill shot

The positive add value comes entirely from **unresolved pullbacks marked at the session close** (optimistic: price drifts back and you flatten at 15:15). We re-priced unresolved adds three ways — **close** (mark at session close), **be** (leg 2 exits flat at its entry, neutral), **stop** (count as a full stop, worst):

Add value (\$) — ALL signals (N=5,444):

PB	close	breakeven	stop (worst)
0.25R	+102,945	–60,746	–782,705

PB	close	breakeven	stop (worst)
0.33R	+60,360	-127,164	-680,190
0.50R	+37,389	-182,418	-494,086
0.66R	-30,821	-206,005	-344,933
0.75R	-38,826	-185,970	-262,201

Add value (\$) — Open+Mid only, the best case (N=3,453):

PB	close	breakeven	stop (worst)
0.25R	+134,012	+19,347	-339,575
0.33R	+109,518	-24,575	-322,357
0.50R	+76,924	-91,458	-279,651
0.66R	+16,701	-118,831	-203,712
0.75R	-2,047	-112,103	-157,909

Add value (\$) — Late+Close (N=1,991):

PB	close	breakeven	stop (worst)
0.25R	-31,068	-80,092	-443,130
0.33R	-49,159	-102,589	-357,834
0.50R	-39,535	-90,960	-214,435
0.66R	-47,521	-87,174	-141,221
0.75R	-36,779	-73,867	-104,292

Strip the optimism and **every configuration goes negative except one** — 0.25R in the morning, at a trivial **+\$19,347 over five years** (~\$6/trade). 50% depth swings from +\$77k to **-\$91k**. Late+Close is negative under every policy and depth.

5. Why it fails

The pullback-to-stop is **more often the start of a loss than a dip to be bought**. On the MC setup, a 50% retrace reaches the original 1R target only ~18% of the time and stops out ~62% of the time — far below the ~27% the second contract needs. So the add is a **negative-expectancy directional bet** that looks profitable only because the backtest marks the leftover unresolved trades at a favorable session close — a drift you can't reliably capture live. Shallower depths don't fix it; they just amplify the same artifact. The math we started with predicted exactly this: a low-hit-rate add against a -1.5R downside can't clear its own bar.

6. Recommendation

Don't scale in. Take the clean single-leg MC trade to 1R and skip the second contract. It needs ~52% wins to break even, the standalone setup is in that range, and you avoid doubling risk into a move that's already against you.

If you scale in anyway: only ever at a **shallow (~0.25R) pullback**, only in the **morning (08:30–13:00 CT)**, and understand you're playing for break-even, not edge.

The one rule worth keeping from all this is negative: avoid the afternoon.

Pullbacks after 13:00 CT are structurally bad — they reach target half as often, every year tested. This applies to the base trade too, not just the scale-in.

7. Caveats & open questions

- **EOD marking is the swing variable.** "Close-mark" is optimistic; "breakeven" is neutral; reality is in between. The ranking (shallow >> deep, morning >> afternoon) is robust; the absolute \$ is not.
- The sim scores trades **independently** (unlimited concurrent positions) — real capital/overlap constraints would only make the marginal add look worse.
- Path levels use raw signal/stop prices; entry slippage is captured via cost, not geometry.
- with_trend (structural-trend agreement) was broken in this run and excluded — worth a clean re-test, though §4.5 makes a strong tell there unlikely.
- Open follow-up: does avoiding the afternoon improve the **base** single-leg MC trade enough to matter? (Suggested by §4.4 but not isolated here.)

8. Reproduce

Scripts (all headless, day-by-day tick replay):

- scripts/pb_to_1r_path_study.py — path buckets + scale-in vs single-leg P&L (§4.2–4.3)
- scripts/pb_tell_hunt.py — TOD, full tell hunt, year split (§4.4–4.6)
- scripts/pb_level_compare.py — pullback-depth sweep, per-CC & per-session (§4.7)
- scripts/pb_eod_stress.py — EOD policy stress test, all scopes (§4.8)
- scripts/render_note_pdf.py <note.md> — renders any note in this series to PDF

Saved artifacts: docs/living/pb_to_1r_paths.parquet, docs/living/pb_tell_features.parquet,
docs/living/pb_level_compare.parquet.